

Planar Transmission Lines

EEL 6486 EM Theory I

Lecture 17

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Don't forget to VOTE

Reviewing The Basics Of Microstrip Lines

An understanding of the fundamentals of microstrip transmission lines can guide high-frequency designers in the proper application of this venerable circuit technology.

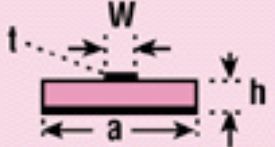
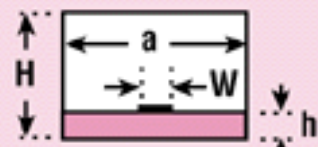
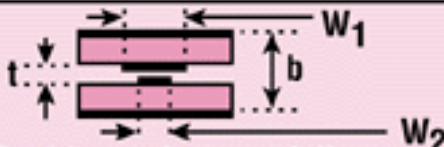
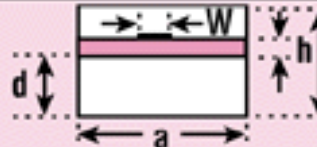
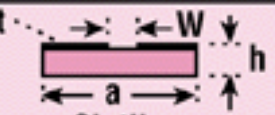
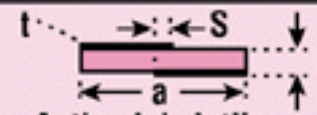
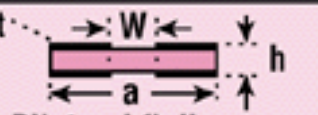
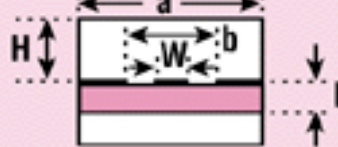
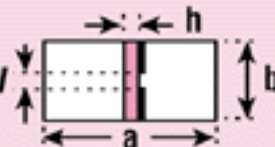
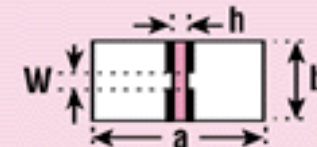
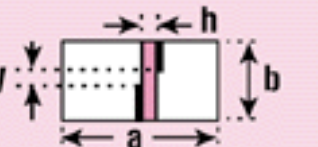
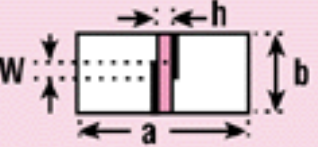
By Leo G. Maloratsky
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[http://www.metcladinternational.com/reference/
Microstrip%20Lines/Microstrip.htm](http://www.metcladinternational.com/reference/Microstrip%20Lines/Microstrip.htm)

(see course website for link)

Why Planar Tx Lines?

- PRINTED transmission lines are widely used, and for good reason. They are broadband in frequency. They provide circuits that are compact and light in weight.
- They are generally economical to produce since they are readily adaptable to hybrid and monolithic integrated-circuit (IC) fabrication technologies at RF and microwave frequencies.
- To better appreciate printed transmission lines, and microstrip in particular, some of the basic principles of microstrip lines will be reviewed here.

	Basic lines	Modifications		
Microstrip line	 Microstrip line	 Suspended microstrip line	 Inverted microstrip line	 Shielded microstrip line
Stripline	 Stripline	 Double-conductor stripline		
Suspended stripline	 Shielded high-Q suspended stripline	 Shielded suspended stripline	 Shielded suspended double-substrate stripline	
Slotline	 Slotline	 Antipodal slotline	 Bilateral finline	
Coplanar waveguide	 Symmetrical coplanar line	 Shielded coplanar waveguide		
Finline	 Finline	 Bilateral slotline	 Antipodal finline	 Antipodal overlapping finline

- In a microstrip line, the guide wavelength, Λ , is given by:

$$\Lambda = \lambda / (\epsilon_{\text{EFF}})^{0.5}$$

where:

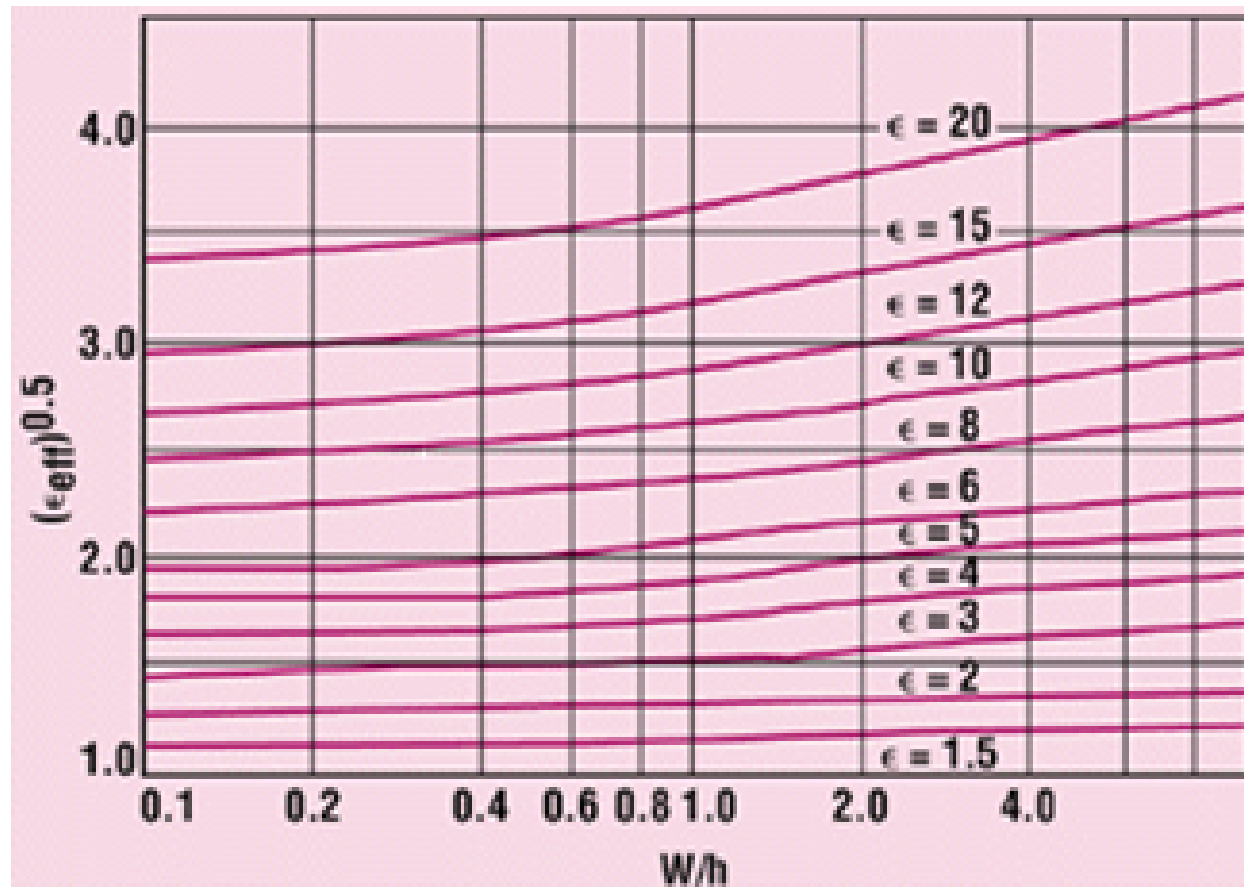
ϵ_{EFF} = the effective dielectric constant,

which depends on the dielectric constant of the substrate material and the physical dimensions of the microstrip line, and

λ = the free-space wavelength.

- In a microstrip line, the electromagnetic (EM) fields exist partly in the air above the dielectric substrate and partly within the substrate itself.
- Intuitively, ϵ_{EFF} of the line is expected to be
- greater than the dielectric constant of air (1) and less than that of the dielectric substrate.

$$\text{i.e., } 1 < \epsilon_{\text{EFF}} < \epsilon_{\text{SUBSTRATE}}$$



Various curves for effective dielectric constant are shown in **Fig. 2** as a function of physical dimensions and relative dielectric constant.

- Referring again to Fig. 1, it should be apparent that a basic (unshielded) microstrip line is not really a practical structure. It is open to the air and, in reality, it is desirable to have circuits that are covered to protect them from the environment as well as to prevent radiation and EM interference (EMI).
- Covering the basic microstrip configuration with metal top plates on the top and on the sides leads to a more realistic circuit configuration, a shielded microstrip line with a housing (Fig. 1).

A comparison of various transmission-line types					
Transmission line	Q factor	Radiation	Dispersion	Impedance range	Chip mounting
Microstrip (dielectric) (GaAs, Si)	250 100 to 150	Low High	Low	20 to 120	Difficult for shunt, easy for series
Stripline	400	Low	None	35 to 250	Poor
Suspended stripline	500	Low	None	40 to 150	Fair
Slotline	100	Medium	High	60 to 200	Easy for shunt, difficult for series
Coplanar waveguide	150	Medium	Low	20 to 250	Easy for series and shunt
Finline	500	None	Low	10 to 400	Fair

The main purposes of the housing or package are to provide mechanical strength, EM shielding, hermetization, and heat sinking in the case of high-power applications.

- An MIC mounted into a housing may be looked on as a dielectrically loaded cavity resonator (**Fig. 3, left**) with the following inner dimensions: a is the width, l is the length, and H is the height of the enclosure.
- These dimensions should be selected in a way so that the waveguide modes are below cutoff.
- Can you think of why this should be so?

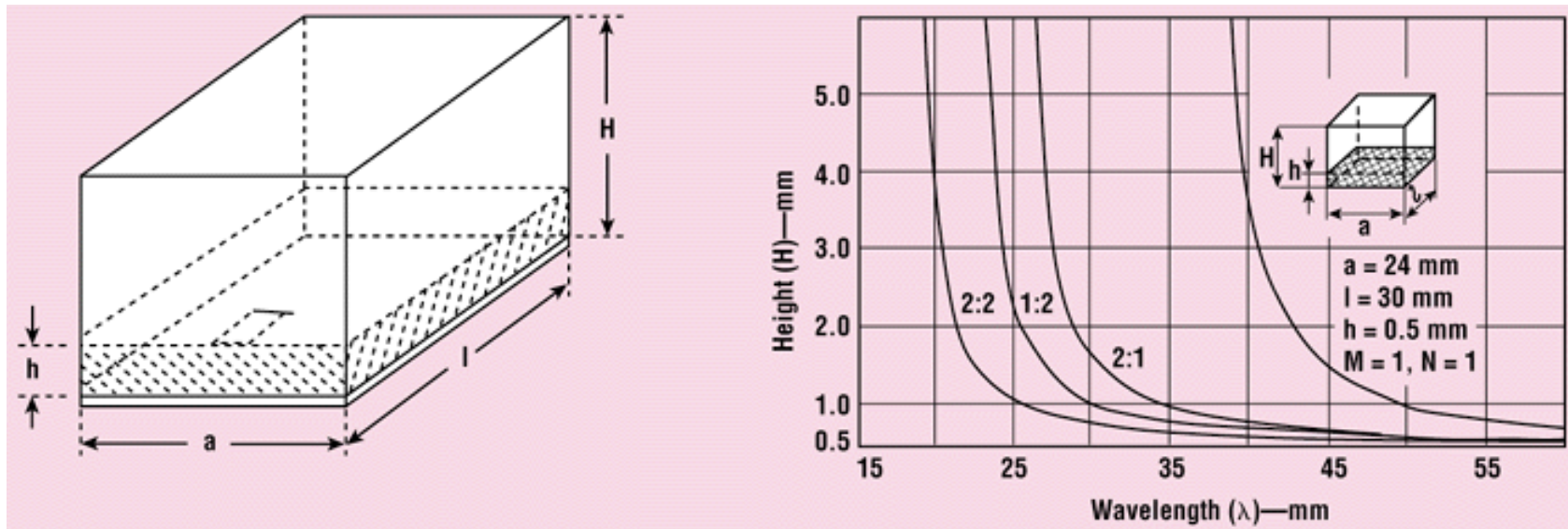


Fig 3. Housing dimensions are selected for microstrip circuits (left) to minimize losses. The effects of unfavorable housing height versus wavelength and different parasitic modes is shown (right).

The parasitic modes appear in this resonator if:

$$H = \{h[1 - (1/\epsilon)]R\}l(R - 1) \quad (2)$$

where:

$$R = (\lambda_0 / 2)^2 [(M / l)^2 + (N / a)^2] \quad (2a)$$

and M and N = positive integers.

From eq. 2, it is possible to obtain the condition of absence of parasitic modes:

$$R - 1 < 0 ; R < 1$$

or

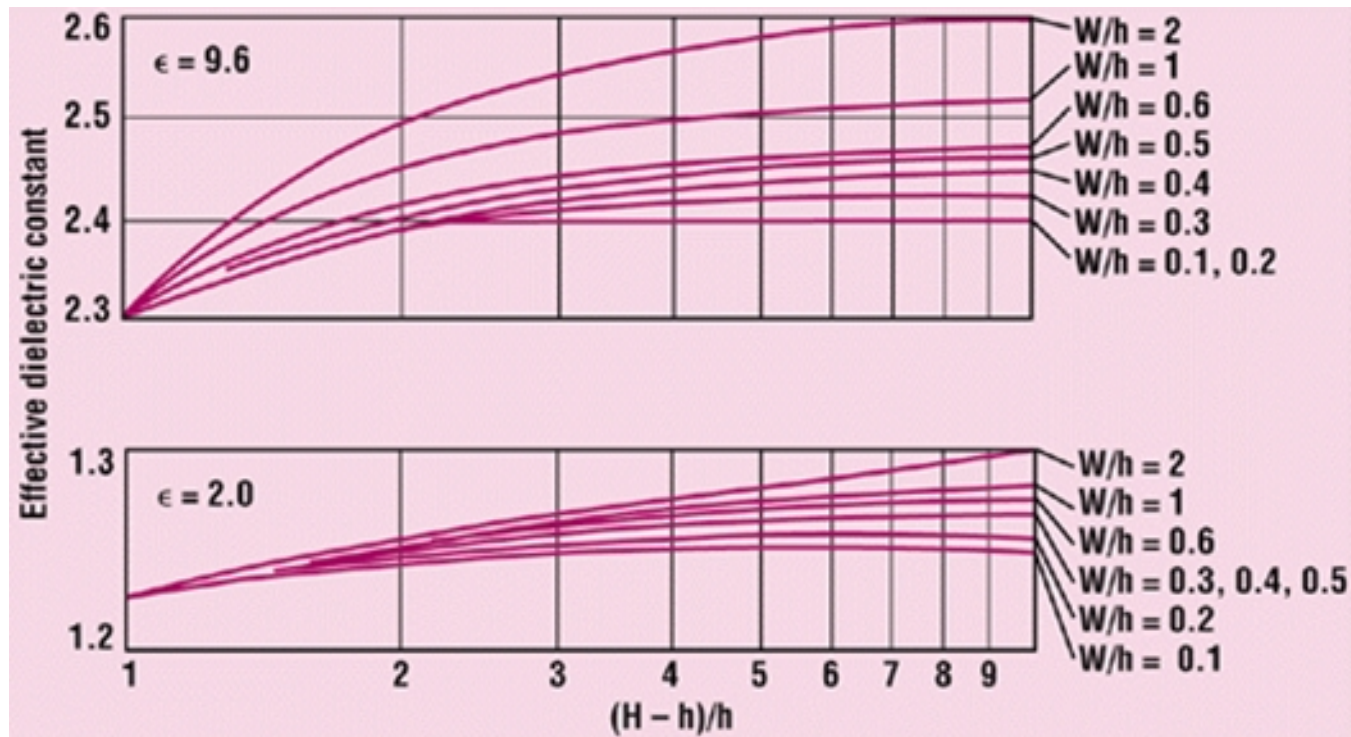
$$\lambda_0^2 < 4 / [(M / l)^2 + (N / a)^2] \quad (3)$$

or

$$\lambda_0 < 2 / [(M / l)^2 + (N / a)^2]^{0.5} \quad (4)$$

- Equation 4 is known as the condition for wave propagation in a waveguide with dimensions $l \times a$. In the case of this article, it can also be considered the condition for the absence of parasitic modes in a waveguide of cross-section $a \times H$ or $l \times H$. If eq. 4 is not satisfied, parasitic modes can arise, and the height H must be chosen to suppress these modes.
- **Figure 3 (right)** illustrates the resulting graphs of unfavorable H versus λ_0 for housing dimensions of $a = 24$ mm, $l = 30$ mm, and dielectric substrate with a dielectric constant of 9.8 and THK of 0.5 mm.

- The top and side covers essentially redistribute the field of the more theoretical microstrip and understandably have an influence on the effective dielectric constant.
- **Figure 4** shows the relationship between the effective dielectric constant and the physical dimensions of the shielded microstrip line for different values of the relative dielectric constant of the substrate material.²



In these curves, it's assumed that the side walls are sufficiently spaced so they only see weak fringing fields. Therefore they have a negligible effect on ϵ_{EFF} .

Top cover tends to lower ϵ_{EFF} (which is consistent with intuition).

The top wall enables electric fields in the air above the strip conductor thereby giving air more influence in determining prop. characteristics.¹⁵

- The characteristic impedance of a microstrip line may be approximately calculated by assuming that the EM field in the line has a quasi transverse-EM (TEM) nature.
- The characteristic impedance of a microstrip line can be calculated using the Wheeler equations.[3](#),[4](#)

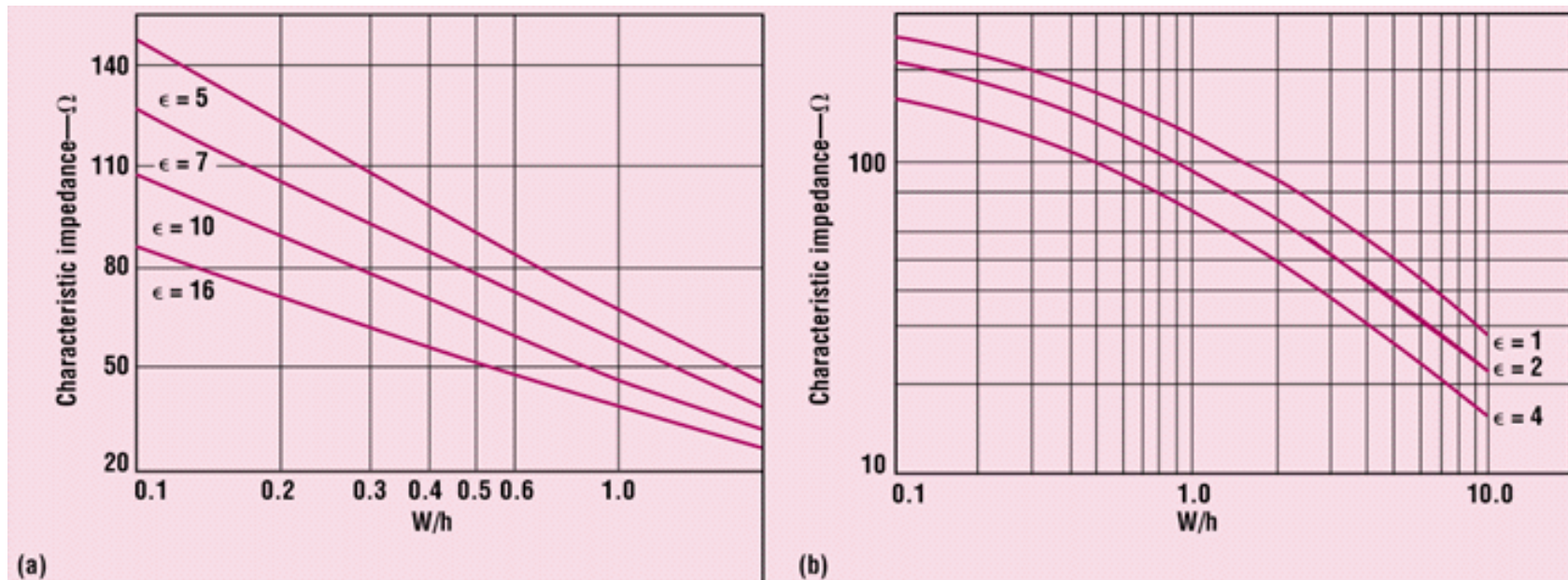


Fig 5. The characteristic line impedance has been plotted for substrates with high (a) and low (b) dielectric constants.

- **Figure 5** shows the characteristic impedance of microstrip lines for various geometries and substrates of different relative dielectric constants.
- **Fig. 6** illustrates the relationships between characteristic impedance and the physical dimensions of shielded microstrip lines for two examples: substrates with low (2) and high (9.6) relative dielectric constants.²

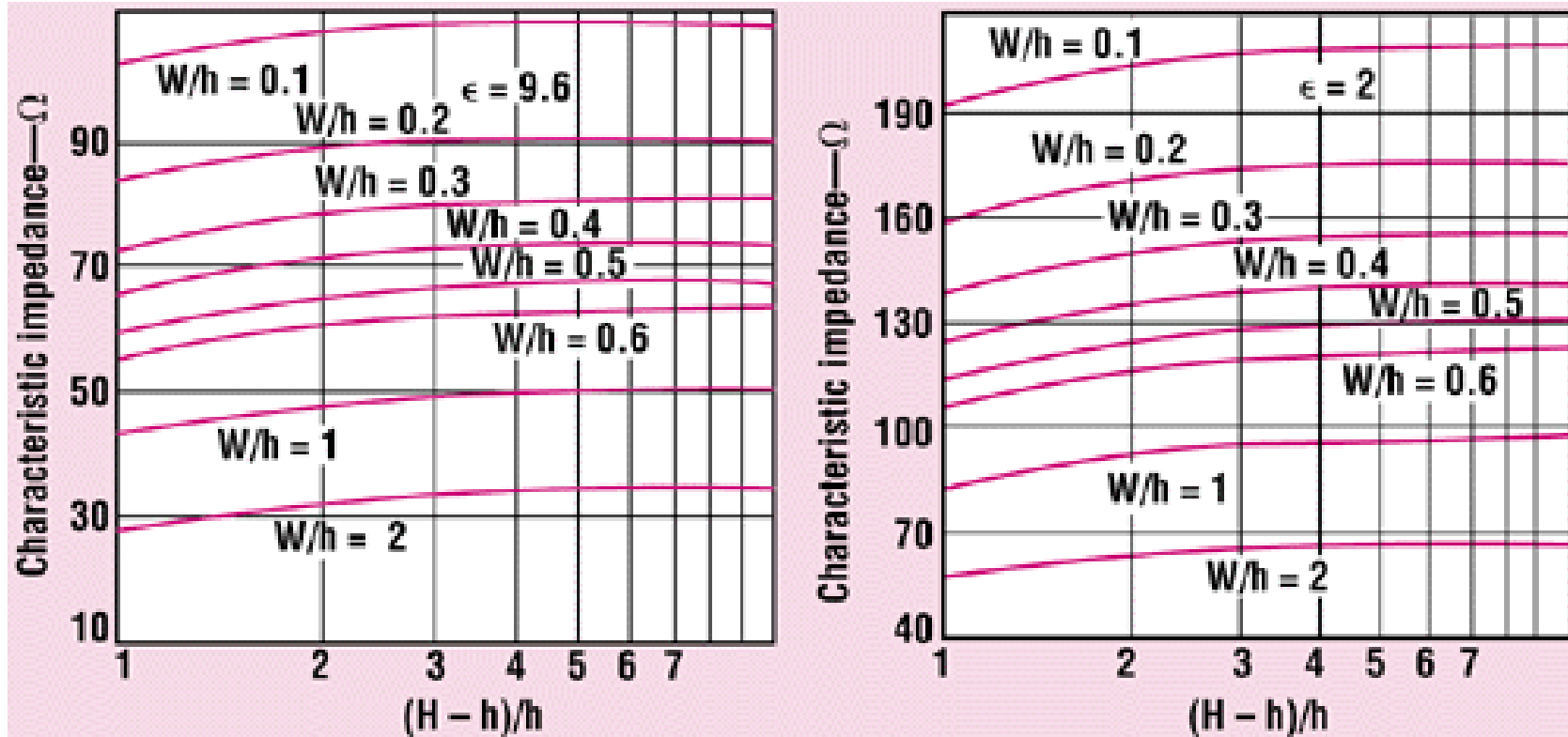
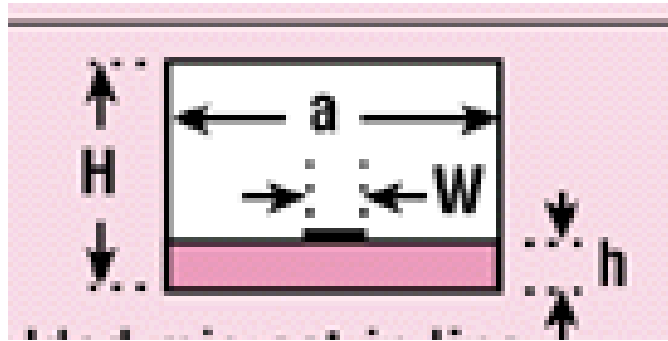


Fig 6. These plots show the relationship between the characteristic impedance and the physical dimensions of microstrip lines using substrates with high (9.6, left) and low (2.0, right) ϵ .



- The top cover tends to reduce the impedance. When the ratio of the distance from the top cover to the dielectric substrate and the substrate thickness $[(H - h)/h]$ is greater than 10, enclosure effects can be considered negligible.
- The characteristic impedance range of a microstrip line is 20 to 120 Ω . The upper limit is set by production tolerances while the lower limit is set by the appearance of higher-order modes.

- Three types of losses that occur in microstrip lines: conductor (ohmic) losses α_C , dielectric losses α_d , and radiation losses α_R .
- Microstrip line, being open to a semi-infinite air space, acts like an *antenna* and tends to *radiate energy*.
- Substrate materials with low ϵ (≤ 5) used when cost reduction is priority. Similar materials used at MMW frequencies to avoid excessively tight mechanical tolerances.
- Lower $\epsilon \Rightarrow$ lower concentration of energy in substrate region, hence $\alpha_R \uparrow$.
- α_R depends on ϵ , the substrate thickness, and the circuit geometry.

- The use of high-dielectric-constant substrate materials reduces radiation losses because most of the EM field is concentrated in the dielectric between the conductive strip and the ground plane.
- The real benefit in having a higher dielectric constant is that the package size decreases by approximately the square root of the dielectric constant (can you tell me WHY?)
- This is an advantage at lower frequencies but may be a problem at higher frequencies.
- If $\tan\delta$ is high then α_R low but α_d high...one of many tradeoffs!

- To minimize $\alpha_C t_{\text{METAL}}$ $3-5 \times$ skin depth, δ .
- In a microstrip line, α_C increases with increasing Z_0 due to the greater resistance of narrow strips.
- α_C follows trend which is *opposite to* α_R with respect to W/h .
- The fabrication process of real microstrip devices creates scratches and bumps on the metal surfaces.
- The current, concentrated metal surface next to substrate, follows uneven surface of substrate and encounters greater resistance compared to case of smooth substrate.
- As roughness of surface increases, current path length increases $\therefore \alpha_C$ increases.

- The rapid development of high-density modules requires the design of interconnects and transitions, especially for multilayer circuits.
- Consider useful transitions from microstrip to other printed transition lines. A transition between two microstrip lines (**Fig. 9a**) can be realized through a slot in the ground plane.

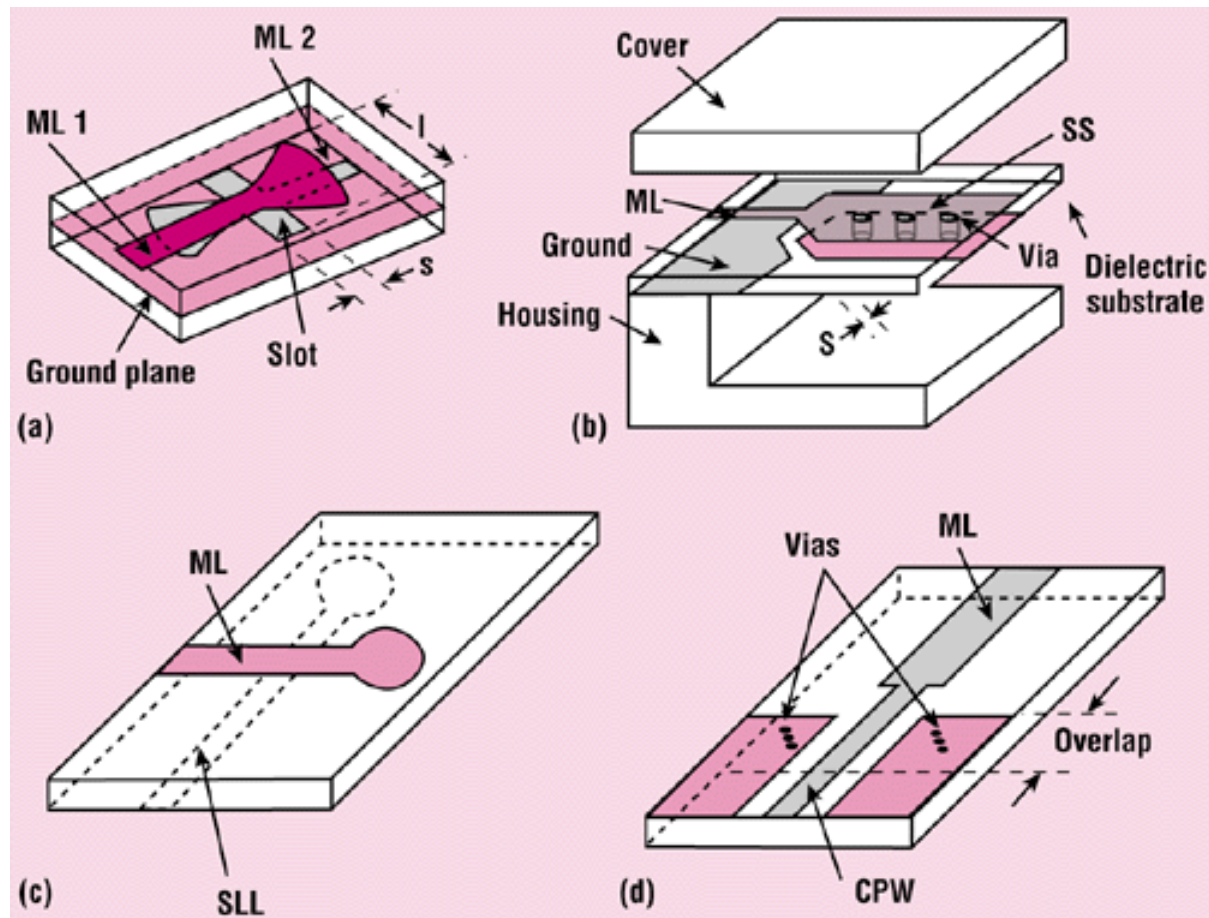


Fig 9. Various transitions between microstrip and other circuit structures are possible: microstrip to microstrip (a), microstrip to suspended stripline (b), microstrip to slotted line (c), and microstrip to coplanar waveguide (d).

- Lets look at some sample hardware
- Then some interesting work done by yours truly over 10 years ago that shows an interesting merger of EM theory and techniques with modern optoelectronics and solid-state devices can be beneficial...